The Coh Category: Verifier-Certified Trace Composition and Projection-Induced Semantic Cost

NoeticanLabs Research Division

April 20, 2026

Abstract

We define the Coh category, a verifier-indexed categorical structure in which morphisms are generated by admissibility rather than assumed a priori. A Coh system consists of observable states, hidden-state realizations, a proposal mechanism, an exact rational valuation, and a deterministic verifier.

Morphisms are finite certified traces satisfying both verifier acceptance and a resource inequality. We prove that these certified traces form a category under concatenation. We then distinguish syntactic bookkeeping cost from semantic cost derived from hidden realizations, and prove a projection-induced subadditive semantic cost law.

A worked example exhibits a strict separation between syntactic and semantic cost, showing that projection can destroy enough hidden information that realized cost is strictly smaller than visible bookkeeping. The result is a precise categorical framework for verifier-certified execution traces in resource-constrained computation.

1 Introduction

In many formal systems, composition is treated as primitive and legality as secondary. In verification-sensitive computation, the order should be reversed: a transition should count as a morphism only after it has survived certification.

The central claim of this paper is therefore:

$$\boxed{\text{Morphisms are verifier-certified traces, not raw arrows.}}$$

This distinguishes the present framework from weighted or resource-aware categories in which morphisms are typically given first and only later decorated with costs; compare the classical enrichment perspective of Kelly [1] and the quantitative viewpoint of Lawvere [2]. In the Coh framework, admissibility generates morphisms.

The scope of the paper is deliberately narrow. We do *not* claim a universal computation substrate. Instead, we isolate a foundational verification wedge: a category of certified execution traces with exact rational bookkeeping and a distinct semantic cost law induced by projection from hidden realizations.

The paper makes four concrete contributions:

- We define a verifier-indexed category whose morphisms are admissible traces.
- We prove categorical closure under trace concatenation.

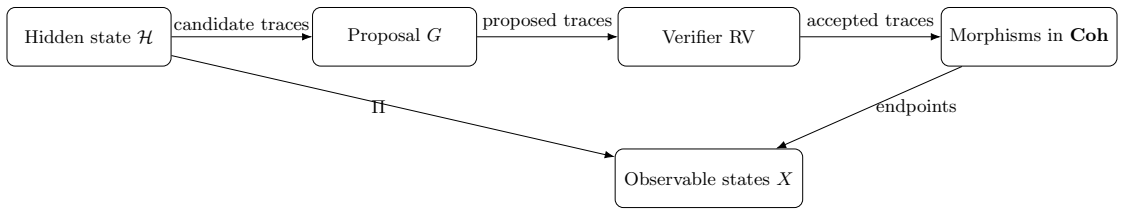


Figure 1: Coh execution wedge. Hidden-state proposals generate candidate traces, the verifier certifies a subset, and only certified traces enter the categorical layer. Projection to observables induces a semantic cost law distinct from syntactic bookkeeping.

- We distinguish syntactic bookkeeping cost from semantic cost and prove a projection-induced subadditive semantic cost law.
- We give a worked example in which semantic cost is strictly smaller than syntactic cost.

2 Preliminaries: exact numerics, payloads, and traces

Definition 2.1 (Exact nonnegative extended rationals)**.** *Define*

$$\overline{\mathbb{Q}}_{\geq 0} := \mathbb{Q}_{\geq 0} \cup \{+\infty\}.$$

The foundational valuation layer of the paper takes values in $\overline{\mathbb{Q}}_{\geq 0}$, while all stepwise bookkeeping quantities take values in $\mathbb{Q}_{\geq 0}$.

Definition 2.2 (Typed payloads and compiled control tokens)**.** *Let X be a set of observable states. For each ordered pair $(x, y) \in X \times X$, let*

$$\mathsf{Payload}(x, y)$$

be a payload type of semantic meaning objects carried by transitions from x to y.

Let Tok be a set of control tokens, and for each (x, y) let

$$\kappa_{x,y} : \mathsf{Payload}(x, y) \to \mathsf{Tok}$$

be a compilation map from payloads to control tokens.

Definition 2.3 (Primitive step)**.** *A primitive step $\sigma : x \rightsquigarrow y$ consists of:*

- *source and target states $x, y \in X$,*
- *a payload $p_\sigma \in \mathsf{Payload}(x, y)$,*
- *a compiled token $\kappa_{x,y}(p_\sigma) \in \mathsf{Tok}$,*
- *exact rational bookkeeping data*
$$s(\sigma), d(\sigma) \in \mathbb{Q}_{\geq 0},$$
- *a well-typedness obligation certifying that the payload is meaningful at (x, y) and compiles to the designated token.*

Definition 2.4 (Finite traces)**.** *Let* $\mathsf{Trace}(X)$ *denote the set of finite composable sequences of primitive steps over* X*. A trace*

$$\tau : x \rightsquigarrow y$$

is written

$$x = x_0 \xrightarrow{\sigma_1} x_1 \xrightarrow{\sigma_2} \cdots \xrightarrow{\sigma_n} x_n = y.$$

Definition 2.5 (Syntactic bookkeeping)**.** *For a trace* τ*, define its syntactic spend and syntactic defect by*

$$s_{\mathrm{syn}}(\tau) := \sum_{i=1}^{n} s(\sigma_i), \qquad d_{\mathrm{syn}}(\tau) := \sum_{i=1}^{n} d(\sigma_i).$$

Definition 2.6 (Concatenation)**.** *If* $\tau_1 : x \rightsquigarrow y$ *and* $\tau_2 : y \rightsquigarrow z$*, define the concatenation*

$$\tau_2 * \tau_1 : x \rightsquigarrow z$$

by appending the step-sequence of τ_2 *after that of* τ_1*.*

3 Coh systems

Definition 3.1 (Coh system)**.** *A Coh system is a tuple*

$$\mathcal{S} = (X, \mathcal{K}, \mathcal{H}, \Pi, \mathsf{Hist}, V, G, \mathrm{RV})$$

where:

- X *is an observable state space,*
- $\mathcal{K} \subseteq X$ *is the admissible subset,*
- $\mathcal{H}$ *is a hidden-state space,*
- $\Pi : \mathcal{H} \to X$ *is the projection to observable state,*
- Hist *is a history/context type,*
- $V : X \to \overline{\mathbb{Q}}_{\geq 0}$ *is a valuation,*
- $G : \mathcal{H} \times \mathsf{Hist} \to \mathcal{P}_{\mathrm{fin}}(\mathsf{Trace}(X))$ *is a proposal mechanism,*
- $\mathrm{RV} : \mathcal{H} \times \mathsf{Trace}(X) \times \mathcal{H} \times \mathsf{Hist} \to \{\mathrm{ACCEPT}, \mathrm{REJECT}\}$ *is a deterministic verifier.*

Definition 3.2 (Finite-valued admissible states)**.** *The admissible subset* $\mathcal{K}$ *satisfies*

$$x \in \mathcal{K} \Longrightarrow V(x) \in \mathbb{Q}_{\geq 0}.$$

Equivalently,

$$\mathcal{K} \subseteq \{x \in X : V(x) < +\infty\}.$$

Definition 3.3 (Hidden fibers)**.** *For* $x \in X$*, define the fiber*

$$\mathcal{F}_x := \{\Psi \in \mathcal{H} : \Pi(\Psi) = x\}.$$

Remark 3.4. *There is exactly one hidden layer in the theory, namely* $\mathcal{H}$*. All hidden representatives are elements of* $\mathcal{H}$*, and all observable states arise by projection through* Π*.*

4 Admissible traces

Definition 4.1 (Admissible trace)**.** *A trace* $\tau : x \rightsquigarrow y$ *is* admissible *if:*

(i) $x, y \in \mathcal{K}$,

(ii) there exist hidden representatives

$$\Psi_x \in \mathcal{F}_x, \qquad \Psi_y \in \mathcal{F}_y,$$

and a history

$$H \in \mathsf{Hist}$$

such that

$$\tau \in G(\Psi_x, H), \qquad \mathrm{RV}(\Psi_x, \tau, \Psi_y, H) = \mathrm{ACCEPT},$$

(iii) the resource inequality holds in $\mathbb{Q}_{\geq 0}$*:*

$$V(y) + s_{\mathrm{syn}}(\tau) \leq V(x) + d_{\mathrm{syn}}(\tau).$$

5 Verifier axioms

Axiom 5.1 (Identity acceptance)**.** *For every* $x \in \mathcal{K}$*, every* $\Psi_x \in \mathcal{F}_x$*, and every admissible history* $H \in \mathsf{Hist}$*, the empty trace*

$$\epsilon_x : x \rightsquigarrow x$$

satisfies

$$\epsilon_x \in G(\Psi_x, H), \qquad \mathrm{RV}(\Psi_x, \epsilon_x, \Psi_x, H) = \mathrm{ACCEPT}.$$

Axiom 5.2 (Verifier closure under concatenation)**.** *Suppose*

$$\tau_1 \in G(\Psi_x, H_1), \qquad \mathrm{RV}(\Psi_x, \tau_1, \Psi_y, H_1) = \mathrm{ACCEPT},$$

and

$$\tau_2 \in G(\Psi_y, H_2), \qquad \mathrm{RV}(\Psi_y, \tau_2, \Psi_z, H_2) = \mathrm{ACCEPT}.$$

Then there exists a compatible history $H_{12} \in \mathsf{Hist}$ *such that*

$$\tau_2 * \tau_1 \in G(\Psi_x, H_{12}), \qquad \mathrm{RV}(\Psi_x, \tau_2 * \tau_1, \Psi_z, H_{12}) = \mathrm{ACCEPT}.$$

Axiom 5.3 (Separation law)**.** *Syntactic spend and syntactic defect are distinct and non-interchangeable:*

$$s_{\mathrm{syn}}(\tau) \not\equiv d_{\mathrm{syn}}(\tau).$$

6 The Coh category

Definition 6.1 (Hom-set)**.** *For* $x, y \in \mathcal{K}$*, define*

$$\mathrm{Hom}_{\mathbf{Coh}(\mathcal{S})}(x, y) := \{\tau : x \rightsquigarrow y \mid \tau \text{ is admissible}\}.$$

Definition 6.2 (Identity morphism)**.** *For each* $x \in \mathcal{K}$*, define*

$$\mathrm{id}_x := \epsilon_x.$$

Definition 6.3 (Composition)**.** *Given*

$$\tau_1 \in \mathrm{Hom}(x, y), \qquad \tau_2 \in \mathrm{Hom}(y, z),$$

define

$$\tau_2 \circ \tau_1 := \tau_2 * \tau_1.$$

Lemma 6.4 (Identity admissibility)**.** *For each $x \in \mathcal{K}$, $\mathrm{id}_x \in \mathrm{Hom}(x, x)$.*

Proof. By Identity Acceptance, the empty trace ϵ_x is verifier-accepted and proposed. Since $x \in \mathcal{K}$, the valuation $V(x)$ is finite rational. Also,

$$s_{\mathrm{syn}}(\epsilon_x) = 0, \qquad d_{\mathrm{syn}}(\epsilon_x) = 0,$$

so the resource inequality reads

$$V(x) + 0 \leq V(x) + 0,$$

which holds. Thus ϵ_x is admissible. □

Lemma 6.5 (Closure under composition)**.** *If*

$$\tau_1 \in \mathrm{Hom}(x, y), \qquad \tau_2 \in \mathrm{Hom}(y, z),$$

then

$$\tau_2 \circ \tau_1 \in \mathrm{Hom}(x, z).$$

Proof. Since τ_1, τ_2 are admissible, we have

$$x, y, z \in \mathcal{K}.$$

Hence

$$V(x), V(y), V(z) \in \mathbb{Q}_{\geq 0}$$

by finite-valued admissibility.

Verifier acceptance of the composite follows from the verifier closure axiom.

For the resource inequality, admissibility gives

$$V(y) + s_{\mathrm{syn}}(\tau_1) \leq V(x) + d_{\mathrm{syn}}(\tau_1),$$

$$V(z) + s_{\mathrm{syn}}(\tau_2) \leq V(y) + d_{\mathrm{syn}}(\tau_2).$$

These are inequalities in $\mathbb{Q}_{\geq 0}$, so ordinary addition is valid. Adding them yields

$$V(y) + V(z) + s_{\mathrm{syn}}(\tau_1) + s_{\mathrm{syn}}(\tau_2) \leq V(x) + V(y) + d_{\mathrm{syn}}(\tau_1) + d_{\mathrm{syn}}(\tau_2).$$

Since $V(y)$ is finite, we may cancel it from both sides and obtain

$$V(z) + s_{\mathrm{syn}}(\tau_1) + s_{\mathrm{syn}}(\tau_2) \leq V(x) + d_{\mathrm{syn}}(\tau_1) + d_{\mathrm{syn}}(\tau_2).$$

Because syntactic bookkeeping is additive under concatenation,

$$s_{\mathrm{syn}}(\tau_2 * \tau_1) = s_{\mathrm{syn}}(\tau_1) + s_{\mathrm{syn}}(\tau_2),$$

$$d_{\mathrm{syn}}(\tau_2 * \tau_1) = d_{\mathrm{syn}}(\tau_1) + d_{\mathrm{syn}}(\tau_2),$$

the concatenated trace satisfies

$$V(z) + s_{\mathrm{syn}}(\tau_2 * \tau_1) \leq V(x) + d_{\mathrm{syn}}(\tau_2 * \tau_1).$$

Thus $\tau_2 * \tau_1$ is admissible. □

$$\begin{tikzcd} x \arrow[r, "\tau_1"] \arrow[rr, bend left, "\tau_2 \circ \tau_1 = \tau_2 * \tau_1"] & y \arrow[r, "\tau_2"] & z \end{tikzcd}$$

Figure 2: Categorical layer. Morphisms are admissible traces, identities are empty traces, and composition is certified trace concatenation.

Theorem 6.6 (Coh category)**.**

$$\mathbf{Coh}(\mathcal{S}) \textit{ is a category.}$$

Proof. Identity morphisms exist by theorem 6.4. Closure under composition holds by theorem 6.5. Associativity follows from associativity of concatenation of finite sequences:

$$(\tau_3 * \tau_2) * \tau_1 = \tau_3 * (\tau_2 * \tau_1).$$

The unit laws follow from the empty trace:

$$\epsilon_y * \tau = \tau, \qquad \tau * \epsilon_x = \tau.$$

□

7 Trace algebra

Theorem 7.1 (Telescoping law)**.** *Let*

$$\tau : x_0 \xrightarrow{\sigma_1} x_1 \xrightarrow{\sigma_2} \cdots \xrightarrow{\sigma_n} x_n$$

be an admissible trace. Then

$$V(x_n) + \sum_{i=1}^{n} s(\sigma_i) \leq V(x_0) + \sum_{i=1}^{n} d(\sigma_i).$$

Proof. By induction on the trace length n.

For $n = 1$, this is exactly the resource inequality in the definition of admissibility.

Assume the result holds for length n. Consider an admissible trace of length $n + 1$:

$$x_0 \xrightarrow{\sigma_1} \cdots \xrightarrow{\sigma_n} x_n \xrightarrow{\sigma_{n+1}} x_{n+1}.$$

By the inductive hypothesis,

$$V(x_n) + \sum_{i=1}^{n} s(\sigma_i) \leq V(x_0) + \sum_{i=1}^{n} d(\sigma_i).$$

By admissibility of the final step,

$$V(x_{n+1}) + s(\sigma_{n+1}) \leq V(x_n) + d(\sigma_{n+1}).$$

Combining the inequalities yields

$$V(x_{n+1}) + \sum_{i=1}^{n+1} s(\sigma_i) \leq V(x_0) + \sum_{i=1}^{n+1} d(\sigma_i).$$

□

Corollary 7.2. *For an admissible trace, intermediate state valuations cancel from the aggregate resource bound.*

8 Semantic cost

Definition 8.1 (Verifier-realizable hidden traces)**.** *For an observable trace τ, let*

$$\mathcal{R}(\tau)$$

denote the finite *set of verifier-realizable hidden traces projecting to τ.*

Definition 8.2 (Hidden trajectory cost)**.** *Let*

$$W : \bigcup_{\tau} \mathcal{R}(\tau) \to \mathbb{Q}_{\geq 0}$$

be a hidden trajectory cost.

Axiom 8.3 (Hidden cost subadditivity)**.** *For composable verifier-realizable hidden traces,*

$$W(\Theta_2 * \Theta_1) \leq W(\Theta_1) + W(\Theta_2).$$

Definition 8.4 (Semantic cost)**.** *For an observable trace τ, define*

$$s_{\mathrm{sem}}(\tau) := \max_{\Theta \in \mathcal{R}(\tau)} W(\Theta).$$

Remark 8.5. *The category uses the additive bookkeeping quantities*

$$s_{\mathrm{syn}},\ d_{\mathrm{syn}},$$

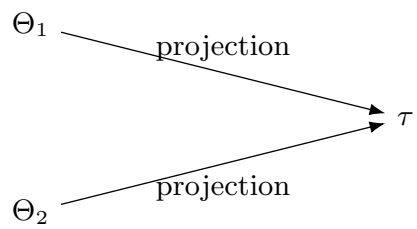
while the semantic theorem below concerns the projected quantity

$$s_{\mathrm{sem}}.$$

This distinction is essential.

Theorem 8.6 (Subadditive semantic cost)**.** *For admissible traces τ_1, τ_2 with compatible endpoints,*

$$s_{\mathrm{sem}}(\tau_2 \circ \tau_1) \leq s_{\mathrm{sem}}(\tau_1) + s_{\mathrm{sem}}(\tau_2).$$



multiple hidden realizations can collapse to one observable trace

Figure 3: Projection from hidden realizations to observables. Multiple hidden traces can collapse to the same observable trace, allowing semantic cost to differ from syntactic bookkeeping.

Proof. By definition,

$$s_{\mathrm{sem}}(\tau_2 \circ \tau_1) = \max_{\Theta \in \mathcal{R}(\tau_2 * \tau_1)} W(\Theta).$$

Each hidden realization

$$\Theta \in \mathcal{R}(\tau_2 * \tau_1)$$

induces compatible restrictions

$$\Theta_1 \in \mathcal{R}(\tau_1), \qquad \Theta_2 \in \mathcal{R}(\tau_2),$$

and hence a restriction map

$$\mathrm{res} : \mathcal{R}(\tau_2 * \tau_1) \to \mathcal{R}(\tau_1) \times \mathcal{R}(\tau_2).$$

By hidden cost subadditivity,

$$W(\Theta) \leq W(\Theta_1) + W(\Theta_2).$$

Taking maxima over the finite sets involved gives

$$\max_{\Theta \in \mathcal{R}(\tau_2 * \tau_1)} W(\Theta) \leq \max_{\Theta_1 \in \mathcal{R}(\tau_1)} W(\Theta_1) + \max_{\Theta_2 \in \mathcal{R}(\tau_2)} W(\Theta_2),$$

which is exactly

$$s_{\mathrm{sem}}(\tau_2 \circ \tau_1) \leq s_{\mathrm{sem}}(\tau_1) + s_{\mathrm{sem}}(\tau_2).$$

□

9 Worked example: semantic cost strictly smaller than syntactic cost

This example is designed to exhibit a genuine separation between syntactic bookkeeping and semantic realized cost.

Definition 9.1 (Toy system)**.** *Let*

$$X = \{A, B, C\}, \qquad \mathcal{K} = X.$$

Let the hidden-state space be

$$\mathcal{H} = \{A_0, A_1, B_0, C_0\},$$

with projection

$$\Pi(A_0) = A, \quad \Pi(A_1) = A, \quad \Pi(B_0) = B, \quad \Pi(C_0) = C.$$

Thus

$$\mathcal{F}_A = \{A_0, A_1\}$$

contains two distinct hidden representatives of the same observable state.

Definition 9.2 (Valuation)**.** *Set*

$$V(A) = 5, \qquad V(B) = 4, \qquad V(C) = 3.$$

Definition 9.3 (Payloads)**.** *Take*

$$\mathsf{Payload}(A, B) = \{\alpha\}, \qquad \mathsf{Payload}(B, C) = \{\beta\}.$$

Let the corresponding control tokens be

$$\kappa_{A,B}(\alpha) = t_\alpha, \qquad \kappa_{B,C}(\beta) = t_\beta.$$

Definition 9.4 (Primitive steps)**.** *Define*

$$\sigma_1 : A \rightsquigarrow B, \qquad \sigma_2 : B \rightsquigarrow C,$$

with bookkeeping data

$$s(\sigma_1) = 1,\ d(\sigma_1) = 0, \qquad s(\sigma_2) = 1,\ d(\sigma_2) = 0.$$

Proposition 9.5. *Both primitive steps satisfy the resource inequality.*

Proof. For σ_1:

$$V(B) + s(\sigma_1) = 4 + 1 = 5 \leq 5 + 0 = V(A) + d(\sigma_1).$$

For σ_2:

$$V(C) + s(\sigma_2) = 3 + 1 = 4 \leq 4 + 0 = V(B) + d(\sigma_2).$$

□

Assume both steps are proposed and verifier-accepted. Let

$$\tau := \sigma_2 * \sigma_1.$$

Definition 9.6 (Composite hidden realization)**.** *Assume*

$$\mathcal{R}(\tau) = \{\Theta^\star\}$$

and

$$W(\Theta^\star) = 1.$$

Intuitively, the composite admits hidden reuse or cancellation not visible in the syntactic bookkeeping.

Proposition 9.7. *For the composite trace τ,*

$$s_{\mathrm{syn}}(\tau) = 2, \qquad s_{\mathrm{sem}}(\tau) = 1.$$

In particular,

$$s_{\mathrm{sem}}(\tau) < s_{\mathrm{syn}}(\tau).$$

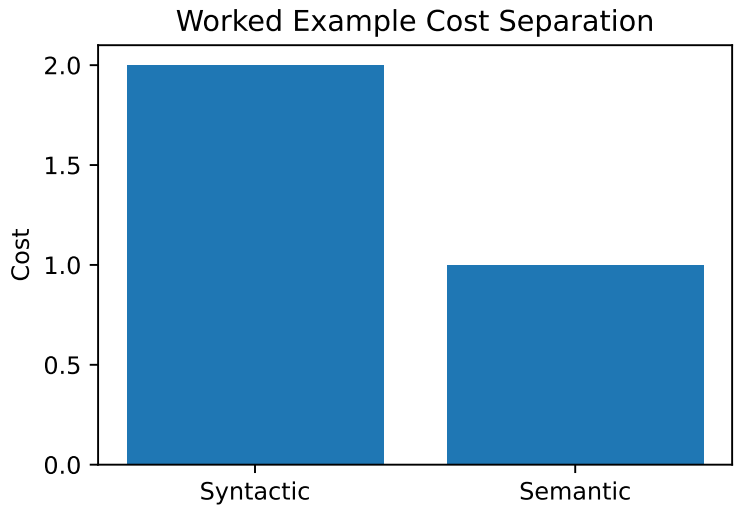


Figure 4: Worked example. Syntactic bookkeeping assigns cost 2 to the composite trace, while semantic cost collapses to 1 after projection through hidden realizations.

Proof. By additivity of syntactic bookkeeping,

$$s_{\mathrm{syn}}(\tau) = s(\sigma_1) + s(\sigma_2) = 1 + 1 = 2.$$

Since $\mathcal{R}(\tau) = \{\Theta^\star\}$, the semantic cost is

$$s_{\mathrm{sem}}(\tau) = \max_{\Theta \in \mathcal{R}(\tau)} W(\Theta) = W(\Theta^\star) = 1.$$

Hence

$$s_{\mathrm{sem}}(\tau) < s_{\mathrm{syn}}(\tau).$$

□

Remark 9.8. *This example shows why the semantic layer is not reducible to bookkeeping. Projection can identify hidden realizations whose actual cost is strictly smaller than the visible syntactic total.*

10 Positioning and scope

The present paper makes a narrow foundational claim.

It is *not* a complete general-purpose computation substrate. It is a formal categorical framework for verifier-certified execution traces with exact rational bookkeeping and a distinct semantic cost law induced by projection.

Its novelty lies in the order of construction:

- traces are first proposed,
- then verified,
- then admitted as morphisms.

Morphisms are therefore not assumed and decorated afterward; they are generated by admissibility.

11 Conclusion

We have defined a category whose morphisms are certified traces rather than freely given arrows. The category theorem depends explicitly on:

- exact rational numerics,
- finite-valued admissible states,
- a proposal mechanism,
- a verifier closed under identity and concatenation,
- additive syntactic bookkeeping.

We then separated syntactic cost from semantic cost and proved that projection from hidden realizations induces a subadditive semantic cost law.

Accordingly, the two main results of the paper are:

$$\boxed{\text{Certified traces form a category under concatenation.}}$$

$$\boxed{\text{Projection can strictly reduce semantic cost relative to syntactic cost.}}$$

This is the correct foundational claim.

A Lean 4 formalization sketch

The sketch below mirrors the paper's semantics more faithfully than placeholder traces of meaningless units. Each step carries a glyph with an explicit compiled control token and proof obligations.

```
inductive GlyphTag
| invoke
| bind
| route
| guard
| emit
deriving DecidableEq, Repr

structure ControlToken where
  opcode : String
  arg    : String
deriving DecidableEq, Repr

structure Glyph where
  tag         : GlyphTag
  surface     : String
  token       : ControlToken
  wf_surface  : Prop
  wf_token    : Prop
```

```
def Glyph.compiles (g : Glyph) : Prop :=
  g.wf_surface and g.wf_token

structure Step (X : Type) where
  src         : X
  dst         : X
  glyph       : Glyph
  costSpend   : Rat
  costDefect  : Rat
  typed       : Prop
  compiles_ok : glyph.compiles

structure Trace (X : Type) where
  steps : List (Step X)
  chain :
    forall  i : Nat, i + 1 < steps.length ->
      (steps.get ⟨i, Nat.lt_trans (Nat.lt_succ_self i) ‹i + 1 < steps.length›⟩).dst =
      (steps.get ⟨i+1, ‹i + 1 < steps.length›⟩).src

def traceSpend {X : Type} (t : Trace X) : Rat :=
  t.steps.foldl (fun acc s => acc + s.costSpend) 0

def traceDefect {X : Type} (t : Trace X) : Rat :=
  t.steps.foldl (fun acc s => acc + s.costDefect) 0

constant RVAccept : {X : Type} -> Trace X -> Prop

def emptyTrace {X : Type} (x : X) : Trace X :=
{ steps := [],
  chain := by
    intro i h
    cases Nat.not_lt_zero _ h }

def concat {X : Type} (t1 t2 : Trace X) : Trace X :=
{ steps := t1.steps ++ t2.steps,
  chain := by
    intro i h
    sorry }

structure CohMor (X : Type) (V : X -> Rat) (x y : X) where
  trace  : Trace X
  rv_ok  : RVAccept trace
  valid  : V y + traceSpend trace <= V x + traceDefect trace

axiom rv_id :
  forall {X : Type} (x : X), RVAccept (emptyTrace x)

axiom rv_comp :
```

```
forall {X : Type} (t1 t2 : Trace X),
  RVAccept t1 -> RVAccept t2 -> RVAccept (concat t1 t2)
```

Remark A.1. *A complete mechanization would refine the concatenation proof for* `chain` *and instantiate the glyph language with a concrete token semantics. The important point for the present paper is that the payload layer is no longer hidden behind* `Type`*: steps now carry explicit meaning objects, compiled control tokens, and proof obligations.*

References

[1] G. M. Kelly, *Basic Concepts of Enriched Category Theory*, Cambridge University Press, 1982.

[2] F. W. Lawvere, Metric spaces, generalized logic, and closed categories, *Rendiconti del Seminario Matematico e Fisico di Milano*, 43 (1973), 135–166.

[3] Jean Bénabou, *Introduction to Bicategories*, Springer, 1967.

[4] Radu Diaconescu, Category-based semantics for equational and constraint logic programming, Technical Report PRG-116, Oxford University Computing Laboratory, 1994.

[5] Joachim Lambek and P. J. Scott, *Introduction to Higher Order Categorical Logic*, Cambridge University Press, 1986.